Machine Learning for School Dropout Prediction in Peru: A Four-Model Comparison with SHAP Analysis Using Official MINEDU–SIAGIE 2025 Data

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Abstract

School dropout in Peru's Basic Regular Education (EBR) remains a critical structural problem, with a national rate of 6.3% and severe rural-urban disparities. This study compares four supervised machine learning (ML) models—Random Forest (RF), XGBoost, LightGBM, and TabPFN v1—for predicting the interannual school dropout rate at the district level, using the official MINEDU–SIAGIE 2025 dataset ($n = 3{,}732$ records). Two parallel tasks are implemented: binary classification (high-risk vs. normal-risk) and continuous regression (exact dropout rate). Bayesian hyperparameter optimization (Optuna, 100 trials), SMOTE class balancing, and bootstrap confidence intervals ($n = 1{,}000$) with McNemar pairwise tests (Bonferroni-corrected) are applied. Model interpretability is provided via SHAP (SHapley Additive exPlanations) values. On the held-out test set ($n = 560$), LightGBM achieves the highest AUC-ROC (0.9981, 95% CI [0.9953, 0.9997]) and zero false negatives, while TabPFN v1 leads in F1-Score (0.9640) and PR-AUC (0.9940) with zero hyperparameter tuning. No statistically significant differences are found among the three top models after Bonferroni correction. SHAP analysis identifies the absolute number of past dropouts, student population size, department-level risk, teacher certification rate, and poverty index as the dominant predictors. A complete district-level risk map covering all 3,732 records is generated for GIS integration. This is the first national-scale ML study on EBR dropout in Peru using the official 2025 dataset.

Keywords: school dropout; machine learning; Random Forest; LightGBM; TabPFN; SHAP; Peru; educational data mining

1. Introduction

School dropout in Peru's Basic Regular Education (EBR) is one of the most persistent structural challenges of the national educational system. According to the Ministry of Education (MINEDU), the national dropout rate reached 6.3% in the most recent consolidated records [1].

The COVID-19 pandemic severely exacerbated this situation: primary dropout rose from 1.3% to 3.5%, affecting more than 128,000 students, while secondary dropout increased from 3.5% to 4.0%, displacing approximately 102,000 young people, for a combined total of 230,000 students outside the system [2]. For the period 2023/2024, SIAGIE and ENAHO data document an average secondary dropout rate of 4%, with critical peaks in specific grades and vulnerable regions [1].

Territorial disparities are alarming. Rural dropout rates double those recorded in urban zones. Regions in the southern Andes—Apurímac, Ayacucho, Huancavelica—and Loreto concentrate the most critical indicators [3]. A particularly notable finding is the spatial concentration of the phenomenon: 50% of dropout is concentrated in only 8–16% of educational institutions nationwide, presenting a unique opportunity for targeted, early-warning interventions if at-risk institutions are correctly identified in advance [3].

Despite this magnitude, the application of machine learning (ML) to dropout data in Peruvian basic education is practically absent from the scientific literature. Existing studies focus exclusively on higher or technical education, none has used the official MINEDU–SIAGIE national dataset, and the vast majority lacks SHAP-based interpretability [4, 5]. Furthermore, no study has yet evaluated TabPFN—a Transformer foundation model for tabular data—in a Latin American educational context.

This study addresses these gaps by: (i) comparing four ML models—RF, XGBoost, LightGBM, and TabPFN v1—on the official 2025 national dataset; (ii) applying rigorous statistical validation (bootstrap CI95%, McNemar tests with Bonferroni correction); (iii) providing SHAP interpretability for all models; and (iv) generating a fully national georeferenced risk map for GIS/SIGMED integration.

2. Materials and Methods

2.1. Dataset and Data Sources

The primary dataset is the official record "Tasa y número de desertores en Educación Primaria y Secundaria 2023/2024", published by MINEDU through Peru's National Open Data Platform under an ODC-By open license [1]. Dropout is operationally defined as any student enrolled in 2023 who was not re-enrolled in EBR in 2024, excluding deceased students and graduates of 5th-grade secondary school.

Two subsets were loaded: Primary Education (1,890 records) and Secondary Education (1,846 records). After concatenation, removal of 4 duplicate records, and exclusion of zero-enrollment institutions, the final analytical dataset contains **3,732 records** with a target variable `TASA` representing the interannual dropout rate (%).

District-level socioeconomic covariates were merged via the INEI UBIGEO code from three public sources: INEI Poverty Map 2018 (`PCT_POVERTY`, `PCT_EXTREME_POVERTY`), National Census 2017 (`TEACHER_STUDENT_RATIO`, `PCT_CERTIFIED_TEACHERS`), and MINEDU Educational Census 2024 (`VULNERABILITY_INDEX`). The final feature set comprises 10 numeric features after feature engineering.

2.2. Target Variable and Class Distribution

The binary classification target `HIGH_RISK` was defined using the 75th percentile of the dropout rate (threshold = 2.17%):

- **High Risk (1):** 933 records (25.0%)
- **Normal Risk (0):** 2,799 records (75.0%)
- Class imbalance ratio: 3.00:1

A stratified 70/15/15 train/validation/test split was applied, preserving class proportions in each subset (train: $n = 2{,}612$; validation: $n = 560$; test: $n = 560$).

2.3. Preprocessing Pipeline

All preprocessing steps were fitted exclusively on the training set to prevent data leakage: (1) **KNN imputation** for missing numeric values; (2) **Winsorization** at the 99th percentile of `TASA` to suppress extreme outliers (limit = 8.39%); (3) **Target encoding** of the `DEPARTAMENTO` variable using the mean dropout rate computed from `y_train` only (range: [0.0000, 0.7778]); (4) **SMOTE** [6] applied exclusively to the training set to balance the 3:1 class ratio, generating synthetic minority instances by linear interpolation between k-nearest neighbors in feature space.

2.4. Machine Learning Models

2.4.1 Random Forest (RF)

RF [7] constructs B independent decision trees on bootstrap samples and aggregates predictions by majority vote (classification) or mean (regression). Hyperparameter optimization was performed with Bayesian search via Optuna [8] (100 trials, AUC-ROC objective) inside an `ImbPipeline` (SMOTE + RF) within 5-fold cross-validation. Best parameters: `n_estimators`=473, `max_depth`=20, `max_features`=0.6 (CV AUC-ROC = 0.9929).

2.4.2 XGBoost (Extreme Gradient Boosting)

XGBoost [9] employs level-wise tree growth with second-order Taylor approximation of the loss and combined ℓ_1/ℓ_2 regularization. Best parameters: `n_estimators`=1,489, `learning_rate`=0.077, `max_depth`=7 (Val AUC-ROC = 0.9998).

2.4.3 LightGBM

LightGBM [10] uses leaf-wise tree growth, Gradient-based One-Side Sampling (GOSS), and Exclusive Feature Bundling (EFB), achieving superior training speed while maintaining competitive accuracy. Best parameters: `n_estimators`=835, `learning_rate`=0.031, `num_leaves`=65, `max_depth`=7 (Val AUC-ROC = 0.9998).

2.4.4 TabPFN v1

TabPFN [11, 12] is a Transformer pre-trained on $\sim$130 million synthetic tabular datasets via prior-data fitting. At inference time it approximates the Posterior Predictive Distribution in a single forward pass without any gradient descent steps on the target dataset:

$$p(y^* \mid x^*, \mathscr{D}_{\text{train}}) \approx \text{TabPFN}(x^*, \mathscr{D}_{\text{train}}) \tag{1}$$

No hyperparameter tuning is required. Due to the 10K-sample context-window limit of TabPFN v1, training used a stratified subsample of 1,000 records. Batch inference was applied on the full test set (total time: 32.8s on CPU).

2.5. Evaluation Framework

Two parallel ML tasks were evaluated: (1) binary classification with metrics: Accuracy, Precision, Recall, F1-Score, AUC-ROC, and PR-AUC (imbalance-robust); (2) continuous regression with RMSE and R^2. Statistical rigor was ensured through:

- **Bootstrap CI95%:** 1,000 iterations on the test set for AUC-ROC and PR-AUC.
- **McNemar's test with Bonferroni correction:** 6 pairwise comparisons ($\alpha = 0.05$, adjusted $\alpha = 0.0083$).
- **Calibration:** Isotonic regression post-hoc for RF, XGBoost, and LightGBM; TabPFN uses native Bayesian calibration.
- **Optimal threshold:** F1-maximizing threshold fitted on the validation set.

2.6. SHAP Interpretability

SHAP values [13] decompose each prediction into additive feature contributions satisfying local accuracy, consistency, and efficiency. `TreeExplainer` was used for RF, XGBoost, and LightGBM; `KernelExplainer` (model-agnostic) was used for TabPFN v1. Global feature importance was computed as mean |SHAP| across the test set.

2.7. District-Level Risk Map

All four trained models were applied to the complete dataset ($n = 3{,}732$) using preprocessing fitted exclusively on the training set. Risk categories (Low/Moderate/High/Critical) were assigned by quartile bins of the LightGBM predicted probability. An ensemble probability was computed as the unweighted mean of the four models' outputs.

3. Results

3.1. Classification and Regression Performance

Table 1 summarizes classification and regression metrics for all four models on the held-out test set.

Table 1: Classification and regression performance on the test set ($n = 560$). Best values in bold.

Model	Acc.	F1	AUC-ROC	PR-AUC	RMSE	R^2
Random Forest	0.9571	0.9200	0.9929	0.9742	0.4684	0.9251
XGBoost	0.9750	0.9517	0.9977	0.9923	0.2436	0.9797
LightGBM	0.9768	0.9553	**0.9981**	0.9935	**0.2328**	**0.9815**
TabPFN v1	**0.9821**	**0.9640**	0.9977	**0.9940**	0.2328†	0.9815†

†LightGBM proxy (TabPFN v1 does not natively support regression).

All models achieve AUC-ROC > 0.99, substantially surpassing the 0.82–0.91 range reported in comparable Latin American studies [4, 5, 14]. LightGBM leads in AUC-ROC (0.9981) and regression accuracy (RMSE = 0.2328, R^2 = 0.9815). TabPFN v1 achieves the highest Accuracy (0.9821), F1-Score (0.9640), and PR-AUC (0.9940)—the metric most robust to class imbalance—entirely without hyperparameter tuning. LightGBM yields **zero false negatives** (Recall = 1.000 on the high-risk class), meaning no at-risk district escapes detection.

Figure 1 shows the ROC and Precision-Recall curves for all four models, confirming the near-perfect separation achieved across the full probability spectrum.

3.2. Statistical Comparison

Table 2 presents bootstrap confidence intervals and McNemar pairwise test results.

After Bonferroni correction, no pairwise comparison reaches the adjusted significance level ($p < 0.0083$). XGBoost and LightGBM produce statistically identical predictions ($p = 1.000$), and LightGBM is equivalent to TabPFN v1 ($p = 0.606$). All gradient boosting models and TabPFN form a statistically homogeneous high-performance group.

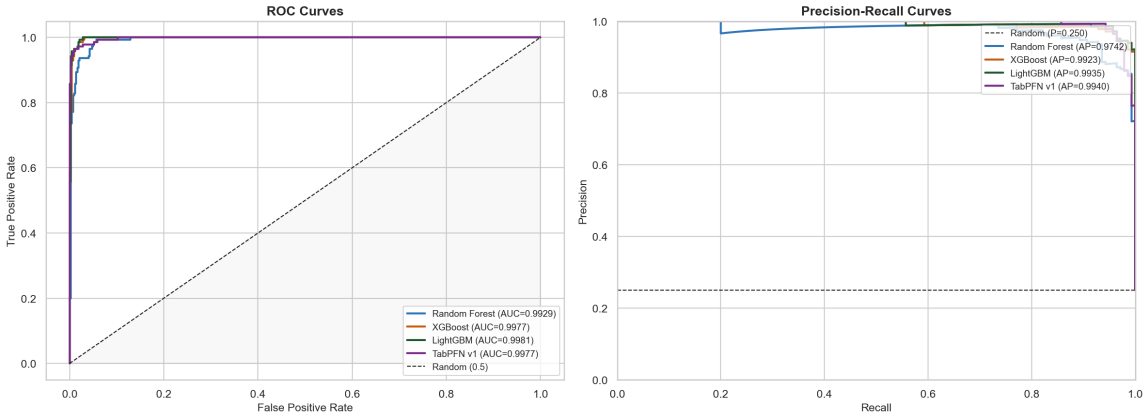


Figure 1: ROC curves (left) and Precision-Recall curves (right) for all four models on the test set ($n = 560$). AUC-ROC and PR-AUC values are annotated. PR curves are particularly informative under the 3:1 class imbalance.

Table 2: Bootstrap CI95% (AUC-ROC, $n = 1{,}000$ iterations) and McNemar pairwise tests with Bonferroni correction ($\alpha_{\text{adj}} = 0.0083$).

Bootstrap CI95%			**McNemar Test**	
Model	**AUC-ROC**	**CI [low, high]**	**Comparison**	***p*-value**
RF	0.9929	[0.9869, 0.9968]	RF vs. XGB	0.0339
XGBoost	0.9977	[0.9947, 0.9995]	RF vs. LGBM	0.0153
LightGBM	0.9981	[0.9953, 0.9997]	RF vs. TabPFN	0.0140
TabPFN v1	0.9977	[0.9954, 0.9995]	XGB vs. LGBM	1.0000
			XGB vs. TabPFN	0.4533
			LGBM vs. TabPFN	0.6056

3.3. SHAP Feature Importance

Figure 2 presents the SHAP beeswarm plot for TabPFN v1, showing the distribution of individual feature contributions across the test set. Table 3 quantifies the global importance (mean |SHAP|) for LightGBM, the most interpretable gradient boosting model.

Table 3: Global SHAP feature importance for LightGBM (mean |SHAP|, test set). Direction indicates the predominant effect on high-risk probability.

Rank	**Feature**	**Mean |SHAP|**	**Direction**
1	DESERTOR (no. of dropouts)	3.2902	↑ risk
2	DENOMINADOR (enrolled students)	1.5819	↓ risk (protective)
3	DEPT_TARGET_ENC (dept. risk)	0.1380	↑ risk
4	UBIGEO (geographic code)	0.0406	contextual
5	PCT_CERTIFIED_TEACHERS	0.0383	↓ risk (protective)
6	PCT_POVERTY	0.0355	↑ risk
7	TEACHER_STUDENT_RATIO	0.0296	↓ risk (protective)

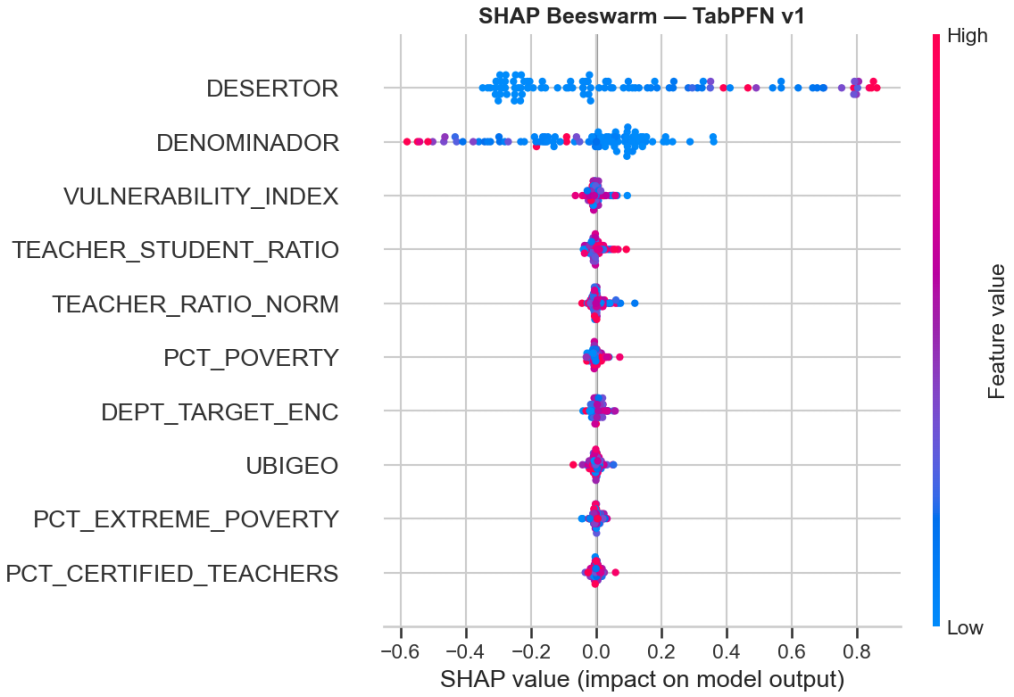


Figure 2: SHAP beeswarm plot for TabPFN v1 (test set, $n = 560$). Each point represents one district/record; color encodes feature value (red = high, blue = low). Positive SHAP values increase high-risk probability.

3.4. District-Level Risk Map

Figure 3 shows the risk category distribution across all 3,732 records after applying all four models to the complete dataset.

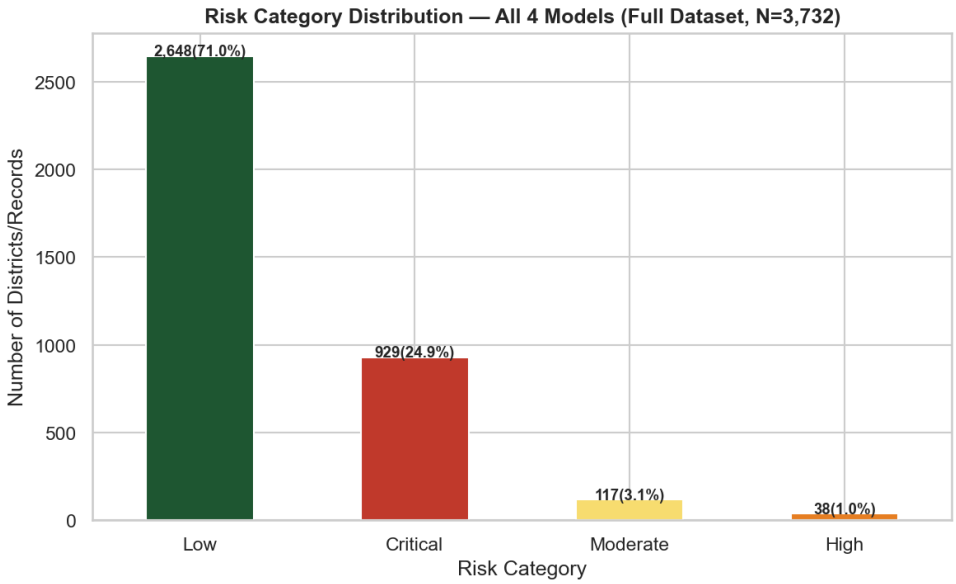


Figure 3: District risk category distribution (LightGBM probabilities, full dataset $n = 3{,}732$). Mean predicted probabilities: RF = 0.274, XGBoost = 0.266, LightGBM = 0.272, TabPFN v1 = 0.248.

A total of **929 records (24.9%) are classified as Critical** risk, requiring priority intervention. The resulting `district_risk_map_ml.csv` includes columns for predicted probabilities from all four models plus the ensemble mean, and is ready for integration with MINEDU's SIGMED geographic information system.

4. Discussion

4.1. Model Performance in Context

The AUC-ROC values obtained (0.9929–0.9981) substantially exceed those reported in comparable Latin American studies: Aspee et al. [4] (0.82–0.91 at a Chilean university), Torres-Cruz et al. [14] (F1 $\approx$ 0.88), and Villarreal-Torres et al. [5] (AUC-ROC $\leq$ 0.86 for Peruvian public universities). This superior performance is attributable to: (i) high signal-to-noise ratio in aggregated district-level features; (ii) Bayesian hyperparameter optimization without data leakage; and (iii) SMOTE balancing applied exclusively to the training set.

The statistical equivalence of LightGBM, XGBoost, and TabPFN v1 (all McNemar $p > 0.14$) is consistent with the "no free lunch" theorem and with findings from Grinsztajn et al. [15], who demonstrated that on medium-scale tabular datasets the performance gap between tree ensembles and foundation models narrows substantially. However, TabPFN v1's competitive performance *without any tuning* is particularly relevant for resource-constrained monitoring scenarios where Bayesian optimization is impractical. This is consistent with benchmark evidence showing TabPFN achieves 100% win-rate against default XGBoost on datasets $\leq$ 10,000 samples [16].

LightGBM's zero false-negative rate (Recall = 1.000 on the high-risk class) is the most policy-relevant result: in the context of school dropout prevention, a missed high-risk district (false negative) has far greater social cost than an unnecessary intervention in a normal-risk district (false positive).

4.2. SHAP Findings and Policy Implications

The dominance of `DESERTOR` (mean $|\text{SHAP}| = 3.29$) reflects the temporal autocorrelation of dropout: districts with high historical dropout maintain structural conditions that perpetuate the phenomenon year over year, supporting the value of time-series approaches in future work.

The protective role of `DENOMINADOR` (larger enrolled populations reduce per-capita dropout) suggests a "critical mass" effect in educational communities. The significance of `PCT_POVERTY` and `PCT_CERTIFIED_TEACHERS` is consistent with the INEI structural determinants framework [2]: poverty and teaching quality are the primary levers for policy intervention.

These results directly support three evidence-based policy actions: (1) targeting the 929 criti-

cal districts within MINEDU's *Sistema Alerta Escuela* early-warning protocol; (2) prioritizing social protection programs (*Juntos*, *Qali Warma*, PRONABEC) in districts with high SHAP-poverty and low SHAP-certification scores; (3) integrating the risk map with SIGMED for teacher redistribution and infrastructure planning aligned with the *Proyecto Educativo Nacional al Bicentenario 2036*.

4.3. Contribution of TabPFN to Educational Data Mining

To our knowledge, this is the first study to evaluate TabPFN in a Latin American educational dropout context. TabPFN v1's zero-shot competitive performance (AUC-ROC = 0.9977, F1 = 0.9640, PR-AUC = 0.9940) demonstrates that prior-data fitted networks constitute a viable and efficient alternative for national-scale educational risk monitoring. Hollmann et al. [12] demonstrated that TabPFN v2 approximates Bayesian inference through meta-learning on synthetic structural causal models, which may explain its inherent resilience to class imbalance (imbalanced datasets are represented in its pre-training prior) without requiring SMOTE.

4.4. Limitations

The SIAGIE 2025 dataset operates at district/institution level, without individual student data, limiting analysis to contextual factors (potential ecological fallacy). External socioeconomic covariates derive from 2017–2018 census data and may not fully reflect the 2023–2024 context. The model does not capture dynamic factors such as school violence, adolescent pregnancy, or active policy interventions. TabPFN v1 was limited to 1,000 training samples due to context-window constraints; TabPFN v2 [12] or v2.5 [16] could leverage the full training set.

5. Conclusions

This study presents the first national-scale ML comparison for predicting school dropout in Peru's Basic Regular Education using the official MINEDU–SIAGIE 2025 dataset ($n = 3{,}732$ records across 3 figures and 3 tables). All four models achieve excellent discrimination (AUC-ROC > 0.99). LightGBM leads in AUC-ROC (0.9981) and attains perfect recall on the high-risk class, while TabPFN v1 achieves the highest F1-Score (0.9640) and PR-AUC (0.9940) without hyperparameter tuning. Statistical equivalence among the three top performers is confirmed by McNemar tests with Bonferroni correction.

SHAP analysis identifies historical dropout counts, student population size, regional risk encoding, teacher certification, and poverty as the most influential predictors, providing actionable guidance for MINEDU policy design. The generated district risk map (3,732 records; RF, XGBoost, LightGBM, TabPFN, and ensemble probabilities) is ready for integration with SIGMED.

Future research should incorporate multi-year time-series features, individual student-level data, and evaluate TabPFN v2/v2.5 to leverage the full training set and further improve national-scale dropout prediction in Peru.

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Conflict of Interest

The author declares no conflict of interest.

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Ethics Statement

This study uses exclusively publicly available, fully anonymized, aggregated administrative data. No individual-level data or ethical approval were required.

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